

Mathematical Analysis Lecture 11 微分学

反函数的求导法则: 若 $y=f(x)$ 在 (a,b) 上连续, 严格单调, 可导, 且 $f'(x) \neq 0$.

记 $\alpha = \min\{f(a+), f(b-)\}$, $\beta = \max\{f(a+), f(b-)\}$. α, β 可以是无穷大.

则反函数 $x=f^{-1}(y)$, $D=(\alpha, \beta)$, 且 $[f^{-1}(y)]' = \frac{1}{f'(x)}$.

$$\text{证: } [f^{-1}(y)]' = \lim_{\Delta y \rightarrow 0} \frac{f^{-1}(y+\Delta y) - f^{-1}(y)}{\Delta y} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{f(x+\Delta x) - f(x)} = \frac{1}{f'(x)}$$

$x=f^{-1}(y)$ 在 (α, β) 上存在, 连续, 严格单调, 则 $\Delta x \neq 0$, $\Delta y \neq 0$, 且 $\Delta x \rightarrow 0$ 时, $\Delta y \rightarrow 0$.

e.g. 求反三角函数的导数: 设 $y = \arcsin x \therefore x = \sin y, y \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

$$\rightarrow \text{微分方法: } dx = \cos y dy \quad \frac{dy}{dx} = \frac{1}{\cos(\arcsin x)} = \frac{1}{\sqrt{1-x^2}}$$

$$\rightarrow \text{反函数求导法则: } (\arcsin x)' = \frac{1}{(\sin y)'} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}} \quad (\arccos x)' = \frac{-1}{\sqrt{1-x^2}} \quad (\arctan x)' = \frac{1}{1+x^2} \quad (\operatorname{arccot} x)' = -\frac{1}{1+x^2}$$

复合函数的求导法则: $u=g(x)$ 在 x 可导, $y=f(u)$ 在 $u=g(x)$ 可导,

则复合函数 $y=f(g(x))$ 在 x 处可导 则 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(u) g'(x)$.

证明略, 已在公式中有所体现。 (下面给出另一种解释, 更严格)

$y=f(u)$ 在 u 可导, 则 $\lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} = f'(u) \therefore \Delta y = f'(u) \Delta u + \alpha \Delta u, (\Delta u \rightarrow 0, \alpha \rightarrow 0)$.

$$\therefore \frac{\Delta y}{\Delta x} = f'(u) \frac{\Delta u}{\Delta x} + \alpha \frac{\Delta u}{\Delta x}, \text{ 取极限: } \frac{dy}{dx} = f'(u) g'(x) \quad (\alpha=0 \text{ 最好})$$

本定理可推广到多个中间变量的情形.

$$\text{e.g. } (x^u)' = (e^{u \ln x})' = e^{u \ln x} \cdot \frac{u}{x} = u x^{u-1}$$

$$\text{e.g. } (x^x)' = (e^{x \ln x})' = e^{x \ln x} (\ln x + 1) = x^x (\ln x + 1)$$

$$\text{可能: } y=x^x, \log_x y = x, \ln y = x \ln x, \therefore \frac{1}{y} dy = (\ln x + 1) dx \therefore \frac{dy}{dx} = x^x (\ln x + 1) \quad (?)$$

$$\text{e.g. } (\sinh x)' = \left(\frac{e^x - e^{-x}}{2} \right)' = \frac{e^x + e^{-x}}{2} = \cosh x. \text{ 同理 } (\cosh x)' = \sinh x.$$

$$\text{e.g. } (\ln \cos(e^x))' = \frac{1}{\cos e^x} \cdot (-\sin e^x) \cdot e^x = -\frac{e^x \sin e^x}{\cos e^x} = -e^x \tan(e^x).$$

$$\operatorname{arsh} x = \ln(x + \sqrt{x^2 + 1}) \quad (\operatorname{arsh} x)' = \frac{1}{\sqrt{1+x^2}} \quad (\text{可用复合法则, 也可以用反函数法则})$$

初等函数的求导法则 (见教材 P.17)

求导善用 $x = e^{\ln x}, a^x = e^{x \ln a}$.

隐函数的导数: 若函数由 $F(x, y) = 0$ 确定, 称为隐函数,

若对 $F(x, y) = 0$ 求导, $\frac{d}{dx} F(x, y) = 0$, 得含有 y' 的方程.

e.g. 对 $F(x, y) = y^5 + 2y - x - 3x^7 = 0$ 求导.

$$\frac{d}{dx} F(x, y) = 0 \Rightarrow 5y^4 \frac{dy}{dx} + 2 \frac{dy}{dx} - 1 - 21x^6 = 0 \therefore y' = \frac{1+21x^6}{5y^4+2}$$

含参方程的导数: 若 $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$ 确定一个函数关系, $\varphi(t), \psi(t)$ 可导, 且二者至少有一个非 0, $\varphi'(t) \neq 0$.

$$\text{所以 } \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\psi'(t)}{\varphi'(t)} \quad \text{含参方程的导数.}$$

Connections?